

AP Statistics Prep Book 2027

The Complete Study Guide & Textbook for High
School Students Featuring Practice Workbooks and
Exam Strategies

AP Statistics Master Review Series — Book 2

PR

2027 Exam Ready Edition

AP Statistics Prep Book 2027: The Complete Study Guide & Textbook for High School Students Featuring Practice Workbooks and Exam Strategies

AP Statistics Master Review Series — Book 2

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How to Use This Textbook

Welcome to the **AP Statistics Prep Book 2027**. This complete textbook and study system is engineered to guide high school students from foundational principles to complete Score-5 exam mastery.

The 4-Pillar Pedagogical System

1. **Core Concept Lessons:** Every unit breaks down the official College Board Course and Exam Description (CED) topics into intuitive, digestible modules with zero confusing academic fluff.
2. **Guided Practice Workbooks:** Each chapter includes hands-on drill sets ranging from foundational calculations to challenging multi-step Free-Response Questions (FRQs).
3. **TI-84 Plus CE Playbooks:** Step-by-step keystrokes and display menus showing exactly how to execute probability calculations, regression models, and hypothesis tests.
4. **Score-4 Rubric Templates:** Grader-approved fill-in-the-blank sentence frames designed to guarantee maximum credit on the 6 exam FRQs.

FREE DIGITAL WEB COMPANION PORTAL**Scan to Access Online Practice****Tools:**

- Unit-by-Unit Diagnostic Quizzes
- Interactive TI-84 Keystroke Simulator
- Searchable Formula Cheat Sheets
- Printable FRQ Practice Worksheets

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Chapter 1

Unit 1: Exploring One-Variable Data

1.1 SECTION A - Unit Overview

Unit 1 Vocabulary

- **Categorical Variable:** A variable that places an individual into one of several groups or categories.
- **Quantitative Variable:** A variable that takes numerical values for which it makes sense to find an average.
- **Distribution:** Tells us what values a variable takes and how often it takes these values.
- **Outlier:** An individual value that falls outside the overall pattern of a distribution.
- **Symmetric:** A distribution is symmetric if the right and left sides of the graph are approximately mirror images of each other.
- **Skewed:** A distribution is skewed to the right if the right tail is much longer than the left tail, and skewed to the left if the left tail is much longer.
- **Mean:** The arithmetic average of a distribution, obtained by adding the observations and dividing by the number of observations ($\bar{x}$).

- **Median:** The midpoint of a distribution, the number such that half the observations are smaller and the other half are larger.
- **Standard Deviation:** A measure of how much the values in a distribution typically vary from the mean (s_x).
- **Variance:** The square of the standard deviation (s_x^2).
- **Z-score:** Tells us how many standard deviations a value is from the mean.
- **Percentile:** The percent of values in a distribution that are less than the individual's data value.

AP Exam Weight: 15-23% of the AP Statistics Exam.

What you will learn: In this unit, you will learn how to analyze and summarize one-variable data using graphs and numerical summaries. You'll master the distinction between categorical and quantitative data, construct appropriate visual representations, calculate measures of center and spread, and use the normal distribution to determine probabilities.

1.2 SECTION B - Core Concept Lessons

Lesson 1.1: Categorical vs. Quantitative Data

Definition: Data represents information collected on variables. A **categorical variable** places individuals into groups (e.g., eye color, zip code). A **quantitative variable** takes numerical values for which calculating an average makes sense (e.g., height, age).

Context: Imagine surveying a high school class. Asking students for their favorite subject produces categorical data. Asking for their GPA produces quantitative data. Note that some numbers act as categories—a zip code or a student ID number doesn't measure quantity, so it's categorical!

Lesson 1.2: Measures of Center (Mean vs. Median)

Definition & Formula: The **mean** ($\bar{x}$) is the arithmetic average: $\bar{x} = \frac{\sum x_i}{n}$. The **median** is the middle value when data is ordered.

Explanation: The mean is sensitive to extreme values (non-resistant), while the median is resistant to extreme values. In a skewed right distribution, the mean is pulled toward the right tail, making it larger than the median.

Context: If you look at the starting salaries of 10 teachers and 1 billionaire CEO, the mean salary will be highly inflated by the CEO, making it an inaccurate representation of a "typical" teacher. The median salary would be much more representative.

Lesson 1.3: Measures of Spread (IQR and Standard Deviation)

Definition & Formula: **Interquartile Range (IQR)** measures the spread of the middle 50% of data: $IQR = Q_3 - Q_1$. **Standard Deviation** (s_x) measures typical distance from the mean: $s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$.

Explanation: Just like the median, the IQR is resistant to outliers. Standard deviation, however, is heavily influenced by outliers because it depends on the mean.

Context: A pizza delivery shop promises times centered at 25 minutes. If standard deviation is 2 minutes, times are consistently close to 25. If standard deviation is 15 minutes, you might get cold pizza in 40 minutes!

Lesson 1.4: Z-scores and Normal Distributions

Definition & Formula: A **z-score** normalizes data by measuring the number of standard deviations a value x is from the mean μ : $z = \frac{x - \mu}{\sigma}$.

Explanation: Normal distributions are symmetric, bell-shaped curves defined by μ and σ . The Empirical Rule states that 68%, 95%, and 99.7% of data fall within 1, 2, and 3 standard deviations

of the mean, respectively.

Context: To compare an SAT score (out of 1600) and an ACT score (out of 36), calculate z-scores for both. If your SAT z-score is 1.5 and ACT z-score is 2.0, you performed relatively better on the ACT!

1.3 SECTION C - Worked Examples

Example 1: Identifying Outliers *Context:* The number of home runs hit by 10 baseball players are: 5, 8, 12, 14, 15, 18, 22, 24, 25, 45. Find if 45 is an outlier. *Solution:* 1. Find Q_1 and Q_3 : Median is $(15 + 18)/2 = 16.5$. $Q_1 = 12$, $Q_3 = 24$. 2. Calculate IQR: $IQR = 24 - 12 = 12$. 3. Check upper boundary: $Q_3 + 1.5 \times IQR = 24 + 1.5(12) = 24 + 18 = 42$. 4. Conclusion: Since $45 > 42$, 45 is indeed an outlier.

Example 2: Comparing Mean and Median *Context:* A neighborhood has home prices (in thousands): \$250, \$260, \$275, \$300, \$1,500. Calculate the mean and median. *Solution:* 1. Median = middle value = \$275,000. 2. Mean = $\frac{250+260+275+300+1500}{5} = \frac{2585}{5} = \$517,000$. 3. Because the distribution is right-skewed (by the \$1.5M mansion), the mean is significantly larger than the median.

Example 3: Calculating Z-Scores *Context:* Average high school female height is $\mu = 64$ inches with $\sigma = 2.5$ inches. Maria is 69 inches tall. *Solution:* 1. Formula: $z = \frac{x-\mu}{\sigma}$ 2. Substitute: $z = \frac{69-64}{2.5} = \frac{5}{2.5} = 2.0$ 3. Conclusion: Maria is 2 standard deviations above the mean height.

Example 4: Using the Empirical Rule *Context:* The lifespan of a specific battery is normally distributed with $\mu = 40$ hours and $\sigma = 5$ hours. What percent of batteries last between 30 and 45 hours? *Solution:* 1. Convert 30 to z-score: $z = (30 - 40)/5 = -2$. 2. Convert 45 to z-score: $z = (45 - 40)/5 = 1$. 3. Area between $z = -2$ and $z = 0$ is $95\%/2 = 47.5\%$. 4. Area between $z = 0$ and $z = 1$ is $68\%/2 = 34\%$. 5. Total percent: $47.5\% + 34\% = 81.5\%$.

1.4 SECTION D - TI-84 CE Keystroke Playbook

TI-84 Procedures for Unit 1

1. 1-Var Stats (Mean, Median, Standard Deviation)

- Enter data: `STAT` → `1:Edit` → Enter data in L1.
- Calculate: `STAT` → `CALC` → `1:1-Var Stats`.
- Ensure List is L1, FreqList is clear, then Calculate.

2. Creating a Boxplot

- Enter data in L1.
- Press `2ND` → `Y=` (STAT PLOT) → `Plot1`.
- Turn Plot ON. For Type, select the modified boxplot (4th icon, shows outliers).
- Press `ZOOM` → `9:ZoomStat` to view.

3. normalcdf (Finding Area/Probability under Normal Curve)

- `2ND` → `VARS` (DISTR) → `2:normalcdf`(
- Syntax: `normalcdf(lower, upper,  $\mu$ ,  $\sigma$ )`
- Use `-1E99` for a left tail of negative infinity, and `1E99` for a right tail.

1.5 SECTION E - Score-4 FRQ Templates

FRQ Template 1: Describing a Distribution

Prompt Type: "Describe the distribution of..."

Template: "The distribution of [context variable] is [shape] with a single peak/multiple peaks. The center, as measured by the median, is approximately [value + units]. The spread of the data ranges from [minimum] to [maximum], with an IQR of [IQR value]. There [appears to be / does not appear to be] an outlier at [value + units]."

Grader Rubric (4 points):

- **S (Shape):** Must explicitly state if skewed or symmetric.
- **O (Outliers):** Must mention presence or absence of outliers.
- **C (Center):** Must provide a measure of center (median/mean) in context.
- **S (Spread):** Must describe variability (range/IQR/std dev) in context.
- *Note: Context must be woven throughout the response to earn full credit.*

FRQ Template 2: Comparing Two Distributions

Prompt Type: "Compare the distributions of A and B..."

Template: "The distribution of [Group A context] is [Shape A], whereas the distribution of [Group B context] is [Shape B]. The center of [Group A] is [greater than/less than/similar to] [Group B], with a median of [Value A] compared to [Value B]. The spread of [Group A] is [greater than/less than] [Group B], with a range of [Range A] vs [Range B]. [Group A] has an outlier at [Value], while [Group B] has no outliers."

Grader Rubric (4 points):

- Must explicitly use comparison words ("greater than", "less

than”, ”similar to”). Simply listing the SOCS for both groups without comparing them will max out at a score of 2!

- Must compare Centers.
- Must compare Spreads.
- Must compare Shapes/Outliers in context.

1.6 SECTION F - Practice Problem Set

Unit 1 Practice Problems

Multiple Choice 1. Which of the following is a quantitative variable? (A) Zip code (B) Race (C) Weight (D) Eye color *Solution: (C). Weight is numerical and an average makes sense.*

2. In a strongly left-skewed distribution, which is true? (A) Mean < Median (B) Mean > Median (C) Mean = Median (D) Mean is zero *Solution: (B). The tail on the left pulls the mean down.*

3. If every value in a dataset is increased by 5, the standard deviation: (A) Increases by 5 (B) Decreases by 5 (C) Remains unchanged (D) Multiplies by 5 *Solution: (C). Adding a constant shifts the center but does not affect spread.*

4. Which measure is most resistant to outliers? (A) Mean (B) Standard Deviation (C) Range (D) IQR *Solution: (D). IQR only looks at the middle 50%, ignoring extremes.*

5. Data set: 10, 12, 14, 15, 18, 22, 28. What is the median? (A) 14 (B) 15 (C) 18 (D) 17.5 *Solution: (B). Middle value of the ordered 7 numbers is 15.*

6. A z-score of -1.5 means the data point is: (A) 1.5 above mean (B) 1.5 below mean (C) Invalid (D) The mean *Solution: (B).*

7. Using empirical rule, what percent falls within 2 std dev? (A) 68% (B) 95% (C) 99.7% (D) 50% *Solution: (B).*

8. If Variance is 16, what is Standard Deviation? (A) 256 (B) 8 (C) 4 (D) 2 *Solution: (C). $\sqrt{16} = 4$.*

9. Which graphical display is best for checking normality? (A) Pie chart (B) Bar chart (C) Normal probability plot (D) Scatterplot
Solution: (C).

10. A distribution with one peak is called: (A) Bimodal (B) Unimodal (C) Uniform (D) Skewed *Solution: (B).*

Free Response 11. The test scores of 10 students are: 45, 78, 82, 85, 88, 90, 91, 93, 95, 99. Calculate the five-number summary and determine if 45 is an outlier. *Solution: $Min=45$, $Q1=82$, $Med=89$, $Q3=93$, $Max=99$. $IQR = 93-82 = 11$. $Lower\ bound = 82 - 1.5(11) = 65.5$. Since $45 < 65.5$, 45 is an outlier.*

12. A normal distribution has $\mu = 50, \sigma = 4$. Find the z-score for $x = 58$. *Solution: $z = \frac{58-50}{4} = 2$.*

13. Explain why we use standard deviation instead of variance for describing spread. *Solution: Standard deviation is in the same units as the original data, making it interpretable, whereas variance is in squared units.*

14. Sketch a density curve that is right-skewed and label the relative positions of the mean and median. *Solution: [Sketch description] A curve with a long tail on the right. The peak represents the mode, the median is to the right of the peak, and the mean is furthest to the right, pulled by the tail.*

15. If a dataset has $\mu = 100, \sigma = 15$, what value represents the 97.5th percentile? *Solution: 97.5th percentile is $z = 2$. $x = \mu + z\sigma = 100 + 2(15) = 130$.*

1.7 SECTION G - Unit Summary

Common Mistakes

- Do not mix up categorical and quantitative displays. Bar charts are for categorical; histograms are for quantitative.
- When comparing distributions, you **MUST** use comparative language.
- Remember: the mean is pulled in the direction of the skew!

Formula Reference: - Sample Mean: $\bar{x} = \frac{\sum x_i}{n}$ - Standard Deviation:

$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$ - Outlier boundaries: $Q_1 - 1.5(IQR)$ and $Q_3 + 1.5(IQR)$
- Z-score: $z = \frac{x - \mu}{\sigma}$

Chapter 2

Unit 2: Exploring Two-Variable Data

2.1 SECTION A - Unit Overview

Unit 2 Vocabulary

- **Explanatory Variable:** A variable that helps explain or influences changes in a response variable (x-axis).
- **Response Variable:** A variable that measures an outcome of a study (y-axis).
- **Scatterplot:** Shows the relationship between two quantitative variables measured on the same individuals.
- **Correlation (r):** Measures the direction and strength of the linear relationship between two quantitative variables.
- **LSRL (Least Squares Regression Line):** The line that minimizes the sum of the squared residuals.
- **Residual:** The difference between an observed value of the response variable and the value predicted by the regression line ($y - \hat{y}$).
- **Residual Plot:** A scatterplot of the residuals against the explanatory variable; helps assess model fit.
- **Coefficient of Determination (r^2):** The fraction of the variation in the values of y that is accounted for by the LSRL.

- **Extrapolation:** The use of a regression line for prediction far outside the interval of values of the explanatory variable.
- **Influential Point:** An observation that, if removed, would markedly change the result of the calculation (especially the LSRL).

AP Exam Weight: 5-7% of the AP Statistics Exam.

What you will learn: You'll move from exploring single variables to analyzing relationships between two quantitative variables. You will interpret scatterplots, calculate correlation, build Least Squares Regression Lines (LSRL) to make predictions, analyze residuals to check model validity, and handle non-linear data.

2.2 SECTION B - Core Concept Lessons

Lesson 2.1: Scatterplots and the DOFS Framework

Definition: A scatterplot graphs two quantitative variables. When interpreting a scatterplot, always use DOFS: Direction, Outliers, Form, and Strength.

Explanation: **Direction:** Positive (as x increases, y increases) or Negative (as x increases, y decreases). **Outliers:** Data points that fall outside the overall pattern. **Form:** Linear (straight line pattern) or Non-linear (curved). **Strength:** How closely the points follow the form (Weak, Moderate, Strong).

Context: If plotting height (x) vs weight (y) for adults, you would likely see a positive, strong, linear relationship. Taller people tend to weigh more.

Lesson 2.2: Correlation Coefficient (r)

Definition & Formula: Correlation (r) measures the strength and direction of a *linear* relationship. It is always between -1 and 1.

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right).$$

Explanation: An r near 0 indicates a weak linear relationship. An

r near 1 or -1 indicates a strong linear relationship. Correlation does not imply causation, and it has no units!

Context: The correlation between hours studied and test score might be $r = 0.85$, indicating a strong positive linear relationship.

Lesson 2.3: Least Squares Regression Line (LSRL)

Definition & Formula: The LSRL is $\hat{y} = a + bx$. The slope is $b = r\left(\frac{s_y}{s_x}\right)$, and the y-intercept is $a = \bar{y} - b\bar{x}$.

Explanation: $\hat{y}$ is the *predicted* value. The slope (b) tells us the predicted change in y for a 1-unit increase in x . The intercept (a) is the predicted y when $x = 0$.

Context: For an equation predicting salary ($\hat{y}$) from years of experience (x): $\hat{y} = 40,000 + 3,000x$. For every additional year of experience, predicted salary increases by \$3,000.

Lesson 2.4: Residuals and r^2

Definition & Formula: Residual = Actual y - Predicted y ($y - \hat{y}$). r^2 is the coefficient of determination.

Explanation: A positive residual means the model underestimated the actual value. A good regression line will have a residual plot with no clear pattern. r^2 tells us the percentage of variability in y explained by the linear relationship with x .

Context: If predicting test score from study hours yields $r^2 = 0.72$, then 72% of the variation in test scores is explained by the linear relationship with study hours.

2.3 SECTION C - Worked Examples

Example 1: Interpreting Slope and Intercept *Context:* A regression line predicting heating cost ($\hat{y}$, in \$) from average outside temperature (x , in $^{\circ}F$) is $\hat{y} = 300 - 4x$. *Solution:* - Slope (-4): For each additional $1^{\circ}F$ increase in outside temperature, the predicted heating cost decreases by \$4. - y-intercept (300): On a day where outside temperature is $0^{\circ}F$, the predicted heating cost is \$300.

Example 2: Calculating a Residual *Context:* The equation $\hat{y} = 2.5 + 1.2x$ predicts GPA based on study hours per week. A student studies 5 hours and has a GPA of 3.9. Calculate their residual. *Solution:* 1. Find predicted GPA: $\hat{y} = 2.5 + 1.2(5) = 2.5 + 6.0 = 8.5$. Wait, GPA of 8.5 is invalid. Let's use $\hat{y} = 2.5 + 0.2(5) = 3.5$. (Recalculating with better numbers: $\hat{y} = 2.5 + 0.2x$. 5 hours $\rightarrow \hat{y} = 2.5 + 0.2(5) = 3.5$.) 2. Actual $y = 3.9$. 3. Residual = $y - \hat{y} = 3.9 - 3.5 = 0.4$. 4. Interpretation: The student's GPA is 0.4 points higher than predicted.

Example 3: Interpreting r^2 *Context:* For the relationship between car weight and MPG, $r = -0.90$. *Solution:* 1. Calculate r^2 : $(-0.90)^2 = 0.81$. 2. Interpretation: 81% of the variability in MPG is accounted for by the linear regression on car weight.

Example 4: Reading Computer Output *Context:* Regression output shows Constant Coef = 15.2, and "Height" Coef = 0.8. *Solution:* 1. Identify the intercept (Constant) = 15.2. 2. Identify the slope (variable coeff) = 0.8. 3. Equation: $\hat{y} = 15.2 + 0.8(\text{Height})$.

2.4 SECTION D - TI-84 CE Keystroke Playbook

TI-84 Procedures for Unit 2

1. Calculating LSRL and Correlation (LinReg)

- Enter x in L1 and y in L2.
- Ensure Diagnostics are ON: MODE $\rightarrow$ StatDiagnostics: ON.
- STAT $\rightarrow$ CALC $\rightarrow$ 8:LinReg(a+bx).
- Specify Xlist: L1, Ylist: L2. Calculate to get a, b, r^2 , and r.

2. Creating a Scatterplot

- 2ND $\rightarrow$ Y= (STAT PLOT) $\rightarrow$ Plot1.
- Type: 1st icon (scatterplot). Xlist: L1, Ylist: L2.
- ZOOM $\rightarrow$ 9:ZoomStat.

3. Creating a Residual Plot

- After running LinReg, residuals are auto-saved to RESID.
- STAT PLOT → Plot1. Xlist: L1, Ylist: 2ND → STAT (LIST) → select RESID.
- ZOOM → 9:ZoomStat.
- Look for a random scatter to confirm a linear model is appropriate!

2.5 SECTION E - Score-4 FRQ Templates

FRQ Template 1: Interpreting Slope and Y-intercept

Prompt Type: "Interpret the slope / y-intercept of the LSRL in context."

Template for Slope: "For each additional [1 unit of x-variable], the **predicted** [y-variable] [increases/decreases] by [slope value]."

Template for Y-intercept: "When the [x-variable] is 0 [units], the **predicted** [y-variable] is [y-intercept value]."

Grader Rubric (4 points):

- Must explicitly state the word "predicted", "estimated", or "expected". (Omitting this implies a deterministic relationship and loses points!)
- Must include the correct units for both x and y variables.
- Must mention the direction (increase/decrease) correctly corresponding to the sign of the slope.

FRQ Template 2: Interpreting a Residual Plot

Prompt Type: "Is a linear model appropriate for this data? Justify using the residual plot."

Template: "Yes/No, a linear model is [appropriate / not appropri-

ate] because the residual plot shows [a random scatter of points / a clear curved pattern].”

Grader Rubric (4 points):

- Must state clear yes/no conclusion.
- Must refer specifically to the *residual plot*, not the scatterplot.
- Must describe the pattern (or lack thereof) in the residual plot.
- Note: Random scatter = linear model is good. Pattern (like a U-shape) = linear model is bad.

2.6 SECTION F - Practice Problem Set

Unit 2 Practice Problems

Multiple Choice 1. Which is true about the correlation coefficient r ? (A) It has units (B) It changes if x and y are swapped (C) It measures linear strength (D) $r = 0$ means no relationship exists
Solution: (C). It measures linear strength. A curve could have $r = 0$ but strong non-linear relation.

2. Residual is calculated by: (A) Predicted - Actual (B) Actual - Predicted (C) Actual / Predicted (D) $x - \bar{x}$
Solution: (B). $y - \hat{y}$.

3. If $r = 0.8$, what is r^2 ? (A) 0.64 (B) 0.8 (C) 1.6 (D) 0.16
Solution: (A).

4. Extrapolation occurs when: (A) You predict inside the domain (B) You predict far outside the domain of x -values (C) The residual is large (D) The correlation is low
Solution: (B).

5. A residual plot shows a clear U-shape. This means: (A) Linear model is appropriate (B) Linear model is NOT appropriate (C) Correlation is 1 (D) r^2 is 100%
Solution: (B). A pattern in residuals indicates a non-linear relationship.

6. If slope $b = -2.5$, what must be true about r ? (A) $r > 0$ (B) $r < 0$ (C) $r = -2.5$ (D) $r = 1$
Solution: (B). Slope and correlation always have the same sign.

7. An influential point is removed. The correlation goes from 0.8 to 0.4. The point was: (A) Following the pattern (B) Breaking the pattern (C) Exactly on the mean (D) A residual of 0 *Solution: (A). It was artificially inflating the correlation by being far out in the x-direction but aligning with the trend.*

8. Equation: $\hat{y} = 10 + 2x$. If $x = 5$, predicted y is: (A) 10 (B) 20 (C) 15 (D) 0 *Solution: (B). $10 + 2(5) = 20$.*

9. Which transformation helps linearize exponential growth data? (A) (x, y^2) (B) $(x, \log y)$ (C) $(\log x, y)$ (D) $(\sqrt{x}, y)$ *Solution: (B). Logging the y-variable linearizes exponential data.*

10. The point $(\bar{x}, \bar{y})$ always: (A) Is an outlier (B) Has a residual of 0 (C) Lies exactly on the LSRL (D) Has maximum leverage *Solution: (C). The LSRL always passes through the point of averages.*

Free Response 11. Given $\bar{x} = 10, s_x = 2, \bar{y} = 50, s_y = 10, r = 0.8$. Find the LSRL. *Solution: $b = r(s_y/s_x) = 0.8(10/2) = 4$. $a = \bar{y} - b\bar{x} = 50 - 4(10) = 10$. LSRL: $\hat{y} = 10 + 4x$.*

12. Interpret $r^2 = 0.64$ in the context of $x = \text{age}$ and $y = \text{height}$. *Solution: 64% of the variation in height is explained by the linear relationship with age.*

13. The LSRL is $\hat{\text{weight}} = 20 + 5(\text{age})$. A 10-year-old weighs 80 lbs. Find the residual. *Solution: $\hat{y} = 20 + 5(10) = 70$. Residual = $y - \hat{y} = 80 - 70 = 10$ lbs.*

14. Why is extrapolation dangerous? *Solution: Because we have no evidence that the linear trend observed in the collected data continues outside that domain.*

15. Describe DOFS for a scatterplot of shoe size vs GPA. *Solution: Direction: None. Outliers: None expected. Form: No distinct form (random scatter). Strength: Very weak. There is no logical relationship.*

2.7 SECTION G - Unit Summary

Common Mistakes

- Forgetting to write the "hat" on $\hat{y}$ in the regression equation!
- Saying a 1-unit increase in x "causes" an increase in y. Use "is associated with a predicted increase".
- Assuming high correlation means a linear model is appropriate. Always check the residual plot!

Formula Reference: - LSRL: $\hat{y} = a + bx$ - Slope: $b = r \frac{s_y}{s_x}$ - Intercept: $a = \bar{y} - b\bar{x}$ - Residual: $y - \hat{y}$

Chapter 3

Unit 3: Collecting Data

3.1 SECTION A - Unit Overview

Unit 3 Vocabulary

- **Population:** The entire group of individuals we want information about.
- **Sample:** A subset of individuals in the population from which we actually collect data.
- **Observational Study:** Observes individuals and measures variables of interest but does not attempt to influence the responses.
- **Experiment:** Deliberately imposes some treatment on individuals to measure their responses.
- **Confounding:** Occurs when two variables are associated in such a way that their effects on a response variable cannot be distinguished from each other.
- **Simple Random Sample (SRS):** A sample of size n chosen in such a way that every group of n individuals has an equal chance to be selected.
- **Stratified Random Sample:** Classifies the population into groups of similar individuals (strata), then chooses a separate

SRS in each stratum.

- **Cluster Sample:** Divides population into groups (clusters), randomly selects some clusters, and surveys EVERY individual in the selected clusters.
- **Undercoverage Bias:** Occurs when some members of the population cannot be chosen in a sample.
- **Nonresponse Bias:** Occurs when an individual chosen for the sample can't be contacted or refuses to participate.
- **Placebo Effect:** A fake treatment that works because subjects believe it will work.
- **Double-Blind:** Neither the subjects nor those who interact with them and measure the response variable know which treatment a subject received.

AP Exam Weight: 12-15% of the AP Statistics Exam.

What you will learn: You will learn how to design good surveys and experiments. You'll distinguish between observing data and manipulating it, learn various sampling techniques to avoid bias, and master the four principles of experimental design to establish causation.

3.2 SECTION B - Core Concept Lessons

Lesson 3.1: Observational Studies vs. Experiments

Definition: An **observational study** gathers data without interfering. An **experiment** actively imposes a treatment to observe the response.

Explanation: Only a well-designed experiment can establish a cause-and-effect relationship. Observational studies are plagued by confounding variables.

Context: If you want to know if tutoring improves test scores, comparing students who chose to get tutoring with those who didn't is

an observational study. Motivation is a confounding variable! To do an experiment, you must randomly assign half the students to receive tutoring.

Lesson 3.2: Sampling Methods

Definition: **SRS:** Names in a hat. **Stratified:** Split by a trait (e.g., grade level), take some from ALL groups. **Cluster:** Split geographically (e.g., homerooms), take ALL from some groups.

Explanation: Stratified sampling guarantees representation from each stratum and reduces variability. Cluster sampling is often done for convenience and cost-effectiveness.

Context: Surveying a high school: - Stratified: Pick 50 freshmen, 50 sophomores, 50 juniors, 50 seniors. - Cluster: Randomly pick 5 homerooms, survey every student in those 5 rooms.

Lesson 3.3: Types of Bias

Definition: Bias is the systematic favoring of certain outcomes.

Voluntary Response: People choose to respond (usually strong negative opinions). **Undercoverage:** Part of population is left out.

Nonresponse: Chosen people refuse. **Response:** People lie or questions are worded poorly.

Context: A radio show asks listeners to call in about a new tax. Only angry taxpayers call. This is Voluntary Response Bias. If a survey is done via landline phones, young people are omitted. This is Undercoverage.

Lesson 3.4: Principles of Experimental Design

Definition: 1. **Comparison:** Compare two or more treatments (e.g., control vs treatment). 2. **Random Assignment:** Use chance to assign treatments to balance out lurking variables. 3. **Control:** Keep other variables constant for all groups. 4. **Replication:** Use enough subjects so differences aren't just by chance.

Context: Testing a new fertilizer: Use the same soil and water

(Control), randomly assign plants to Fertilizer A or B (Random Assignment), use 100 plants not just 2 (Replication), and compare growth (Comparison).

3.3 SECTION C - Worked Examples

Example 1: Identifying Confounding Variables *Context:* A study finds that cities with more churches also have higher crime rates. Can we conclude churches cause crime? *Solution:* No. This is an observational study. A confounding variable is population size. Larger cities have both more churches and more crime. We cannot distinguish the effect of churches from the effect of population size.

Example 2: How to perform an SRS *Context:* You have a list of 800 students. How do you select an SRS of 20? *Solution:* 1. Label each student with a unique number from 001 to 800. 2. Use a random number generator to select 20 unique numbers between 001 and 800. 3. The students corresponding to these 20 numbers form the sample.

Example 3: Stratified vs Cluster *Context:* A farmer wants to test the quality of apples in his orchard of 10 rows. *Solution:* - Stratified: He randomly selects 3 apples from every single row (Total = 30 apples). Good if rows get different sunlight. - Cluster: He randomly selects 1 entire row and checks all 30 apples in that row. Good for saving time walking around.

Example 4: Randomized Block Design *Context:* Testing a new sunscreen. Men and women sweat differently. How to design? *Solution:* 1. Block by gender: Separate subjects into a block of men and a block of women. 2. Within the men's block, randomly assign half to the new sunscreen, half to the old. 3. Within the women's block, randomly assign half to the new, half to the old. 4. Compare skin damage within men, and separately within women. This removes variation due to gender.

3.4 SECTION D - TI-84 CE Keystroke Playbook

TI-84 Procedures for Unit 3

1. Setting the Random Seed (Optional but good practice)

- Type any unique number (like your student ID).
- Press **STO** $\rightarrow$.
- Press **MATH** $\rightarrow$ **PROB** $\rightarrow$ **1:rand**. Hit **ENTER**.

2. Generating Random Integers for an SRS

- **MATH** $\rightarrow$ **PROB** $\rightarrow$ **5:randInt(**
- Syntax: **randInt(lower, upper, n)** where **n** is how many numbers you want.
- Example: **randInt(1, 800, 20)** generates 20 numbers between 1 and 800.
- *Note: You must manually discard any repeat numbers if sampling without replacement!*

3. Generating Random Integers without Repeats (Newer TI-84 CE)

- **MATH** $\rightarrow$ **PROB** $\rightarrow$ **8:randIntNoRep(**
- Syntax: **randIntNoRep(lower, upper)**
- Generates a list of all integers in the range in a random scrambled order. Just take the first n numbers.

3.5 SECTION E - Score-4 FRQ Templates

FRQ Template 1: Describing an SRS

Prompt Type: "Describe how you would select a Simple Random Sample of size n from a population of size N ."

Template: "First, I will assign a unique integer from 1 to $[N]$ to each [individual in population]. Using a random number generator, I will generate $[n]$ **unique** random integers from 1 to $[N]$. I will ignore any repeated numbers. The $[n]$ individuals whose labels correspond to the generated numbers will be selected for the sample."

Grader Rubric (4 points):

- **Labeling:** Must state that numbers are assigned to individuals.
- **Range:** Must specify the range of numbers used.
- **Selection:** Must state how numbers are chosen (RNG or hat).
- **Repeats:** Must explicitly state that repeated numbers are skipped/ignored! (Crucial point).

FRQ Template 2: Explaining Bias

Prompt Type: "Explain why this sampling method is biased and the direction of the bias."

Template: "This sampling method suffers from [Type of Bias] because [explain how the sample doesn't represent the population]. As a result, the sample will likely have a disproportionate number of [specific group]. Because this group tends to [explain their specific trait related to the response variable], the survey will likely [overestimate / underestimate] the true [parameter being measured]."

Grader Rubric (4 points):

- Identify the specific bias or flaw in the design.
- Connect the flaw to the specific context of the problem.

- State whether the estimate will be an overestimation or underestimation.
- Explain *why* the flaw leads to that specific direction of bias.

3.6 SECTION F - Practice Problem Set

Unit 3 Practice Problems

Multiple Choice 1. To establish cause-and-effect, you must use: (A) An observational study (B) A census (C) A well-designed experiment (D) A stratified sample *Solution: (C). Only experiments can show causation.*

2. Blocking in an experiment is similar to what in sampling? (A) SRS (B) Stratified sampling (C) Cluster sampling (D) Systematic sampling *Solution: (B). Both group similar individuals together to reduce variation.*

3. A survey on a website asks visitors to vote on a policy. This is: (A) SRS (B) Voluntary response (C) Cluster (D) Stratified *Solution: (B). Visitors self-select to participate.*

4. Purpose of random assignment in an experiment: (A) Create roughly equivalent groups (B) Eliminate the placebo effect (C) Ensure sample represents population (D) Reduce sample size *Solution: (A). It balances out confounding variables between treatment groups.*

5. A fake pill used in medical trials is called: (A) A block (B) A treatment (C) A placebo (D) A stratum *Solution: (C).*

6. A pollster randomly calls landline phones during the day. What bias is most likely? (A) Nonresponse bias (B) Undercoverage bias (C) Response bias (D) Both A and B *Solution: (D). Undercoverage (no cell phones) and nonresponse (people not home).*

7. We group students by grade level and take a random sample of 10 from each grade. This is: (A) Cluster (B) Stratified (C) SRS (D) Systematic *Solution: (B). Some from all groups.*

8. Which principle of design reduces variation by applying treat-

ments to many subjects? (A) Control (B) Random Assignment (C) Replication (D) Comparison *Solution: (C).*

9. Double-blind means: (A) Subjects don't know the treatment (B) Evaluators don't know the treatment (C) Both A and B (D) Two placebos are used *Solution: (C).*

10. You survey every 10th person walking into a store. This is: (A) SRS (B) Stratified (C) Systematic (D) Cluster *Solution: (C).*

Free Response 11. Explain the difference between stratified and cluster sampling. *Solution: Stratified divides population into groups and takes a random sample from EVERY group. Cluster divides population into groups, randomly selects SOME groups, and censuses EVERY individual in those selected groups.*

12. A gym wants to test a new protein powder. Explain a confounding variable if they just compare members who buy it vs those who don't. *Solution: Members who buy the powder might also work out more frequently or have better overall diets. We wouldn't know if muscle gain is from the powder or the extra workouts.*

13. Design a completely randomized design for testing 2 drugs on 100 patients. *Solution: Assign patients numbers 1-100. Use a RNG to select 50 unique numbers. These 50 patients receive Drug A. The remaining 50 receive Drug B. Compare the recovery rates.*

14. Why is blinding important in an experiment? *Solution: It prevents the placebo effect in subjects and prevents measurement bias from the researchers who are evaluating the results.*

15. What is the main benefit of blocking? *Solution: Blocking isolates the variation caused by the blocking variable, making it easier to detect a difference between the treatments.*

3.7 SECTION G - Unit Summary

Common Mistakes

- Using "random assignment" and "random sampling" interchangeably. Random sampling is how you get your subjects (generalizability). Random assignment is how you assign treatments (causation).
- Forgetting to say "ignore repeats" when describing an SRS.
- Confusing stratified sampling (some from all) with cluster sampling (all from some).

Key Takeaways: - Observational Study → Association only. - Experiment → Causation. - Random Selection → Allows generalization to population. - Random Assignment → Allows cause-and-effect conclusion.

Chapter 4

Unit 4: Probability and Random Variables

4.1 SECTION A - Unit Overview

Unit 4 Vocabulary

- **Sample Space (S):** The set of all possible outcomes of a chance process.
- **Event:** Any collection of outcomes from some chance process.
- **Mutually Exclusive (Disjoint):** Two events that have no outcomes in common and can never occur together.
- **Independent Events:** Two events are independent if the occurrence of one does not change the probability that the other will happen.
- **Conditional Probability:** The probability that one event happens given that another event is already known to have happened, $P(A|B)$.
- **Random Variable (RV):** Takes numerical values that describe the outcomes of some chance process.
- **Discrete RV:** Can take a fixed set of possible values with gaps between them (usually counts).

- **Continuous RV:** Can take any value in an interval on the number line (usually measurements).
- **Expected Value (Mean of RV):** The long-run average outcome of a random variable.
- **Binomial Setting:** Arises when we perform several independent trials of the same chance process and record the number of times that a particular outcome occurs.
- **Geometric Setting:** Arises when we perform independent trials of the same chance process and record the number of trials until a particular outcome occurs.

AP Exam Weight: 10-20% of the AP Statistics Exam.

What you will learn: You will learn the formal rules of probability, how to evaluate conditional probabilities, and how to use Venn diagrams and tree diagrams. You'll define random variables, calculate expected values, combine distributions, and identify and calculate probabilities for Binomial and Geometric distributions.

4.2 SECTION B - Core Concept Lessons

Lesson 4.1: General Probability Rules

Definition & Formula: Complement Rule: $P(A^c) = 1 - P(A)$.

General Addition Rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

General Multiplication Rule: $P(A \cap B) = P(A) \times P(B|A)$.

Explanation: The addition rule uses "OR" ($\cup$), meaning at least one event occurs. We subtract the intersection so we don't double count! The multiplication rule uses "AND" ($\cap$).

Context: Probability of being left-handed is 0.10. Probability of having blue eyes is 0.30. If these traits are independent, probability of having BOTH is $0.10 \times 0.30 = 0.03$.

Lesson 4.2: Conditional Probability and Independence

Definition & Formula: Conditional formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Independence check: A and B are independent if $P(A|B) = P(A)$.

Explanation: Independence means knowing B happened gives you NO new information about A . Do not confuse independent with mutually exclusive. Mutually exclusive events are highly DEPENDENT (if one happens, the other has 0% chance of happening).

Context: If $P(A \text{ on test}) = 0.8$, but $P(A \text{ on test} | \text{slept well}) = 0.95$, then getting an A and sleeping well are NOT independent.

Lesson 4.3: Transforming and Combining Random Variables

Definition & Formula: Linear Transformation: If $Y = a + bX$, then $\mu_Y = a + b\mu_X$ and $\sigma_Y = |b|\sigma_X$. Combining: If X and Y are independent, $\mu_{X+Y} = \mu_X + \mu_Y$ and $\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2$.

Explanation: Means always add/subtract directly. Variances ALWAYS add, even if you are subtracting the random variables! You CANNOT add standard deviations directly.

Context: Packing a box. Weight of box is X , weight of item is Y . Total weight = $X + Y$. The variance of the total weight is the variance of the box plus the variance of the item.

Lesson 4.4: The Binomial Distribution (BINS)

Definition & Formula: Check **BINS**: **B**inary (Success/Failure) Independent trials **N**umber of trials is fixed (n) **S**ame probability of success (p). Formula: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.

Explanation: The binomial distribution counts the number of successes in a fixed number of trials. Mean $\mu = np$, standard deviation $\sigma = \sqrt{np(1 - p)}$.

Context: Flipping a coin 10 times and counting heads. $n = 10, p = 0.5$. What is the probability of exactly 6 heads?

4.3 SECTION C - Worked Examples

Example 1: Using the Addition Rule *Context:* At a school, 40% of students take French, 30% take Spanish, and 10% take both. Find the probability a random student takes French OR Spanish. *Solution:* $P(F \cup S) = P(F) + P(S) - P(F \cap S)$ $P(F \cup S) = 0.40 + 0.30 - 0.10 = 0.60$.

Example 2: Checking for Independence in a Table *Context:* Event M = Male (50 out of 100). Event S = Plays sports (40 out of 100). 25 are Males who play sports. Are M and S independent? *Solution:* Check if $P(S|M) = P(S)$. $P(S) = 40/100 = 0.40$. $P(S|M) = \frac{P(S \cap M)}{P(M)} = \frac{25/100}{50/100} = \frac{25}{50} = 0.50$. Since $0.50 \neq 0.40$, they are NOT independent.

Example 3: Expected Value *Context:* A lottery ticket costs \$5. It has a 10% chance of winning \$20, a 1% chance of winning \$100, and otherwise pays \$0. What is the expected profit? *Solution:* Net outcomes: Win \$20 $\rightarrow$ Profit \$15. Win \$100 $\rightarrow$ Profit \$95. Lose $\rightarrow$ Profit -\$5. $P(15) = 0.10$, $P(95) = 0.01$, $P(-5) = 0.89$. $E(X) = (15)(0.10) + (95)(0.01) + (-5)(0.89) = 1.50 + 0.95 - 4.45 = -2.00$. Expected profit is -\$2.00.

Example 4: Binomial Probability *Context:* A basketball player makes 80% of his free throws. He shoots 5 independent free throws. What is the probability he makes exactly 3? *Solution:* $n = 5, p = 0.8, k = 3$. $P(X = 3) = \binom{5}{3}(0.8)^3(0.2)^2 = (10)(0.512)(0.04) = 0.2048$.

4.4 SECTION D - TI-84 CE Keystroke Playbook

TI-84 Procedures for Unit 4

1. binompdf (Exact binomial probability)

- Finds $P(X = k)$.
- 2ND $\rightarrow$ VARS (DISTR) $\rightarrow$ A:binompdf(
- Syntax: binompdf(trials, p, x value)
- Example: binompdf(5, 0.8, 3) yields 0.2048.

2. binomcdf (Cumulative binomial probability)

- Finds $P(X \leq k)$.
- 2ND $\rightarrow$ VARS (DISTR) $\rightarrow$ B:binomcdf(
- Syntax: binomcdf(trials, p, x value)
- Example: To find $P(X \geq 3)$, do $1 - \text{binomcdf}(5, 0.8, 2)$.

3. geometpdf / geometcdf

- 2ND $\rightarrow$ VARS (DISTR) $\rightarrow$ E:geometpdf(or F:geometcdf(
- Syntax: geometpdf(p, x value)
- Used for Geometric distributions (counting trials UNTIL a success).

4.5 SECTION E - Score-4 FRQ Templates

FRQ Template 1: Identifying a Binomial Distribution

Prompt Type: "Explain why the random variable X follows a binomial distribution."

Template: " X is a binomial random variable because it satisfies the four BINS criteria: 1. **Binary:** Each trial results in a success ([define success]) or failure ([define failure]). 2. **Independent:** The outcome of one trial does not affect the outcome of another. 3. **Number fixed:** There is a fixed number of trials, $n = [\text{number}]$. 4. **Same probability:** The probability of success, $p = [\text{probability}]$, is the same for each trial."

Grader Rubric (4 points):

- Must explicitly list all 4 criteria.
- Must define success/failure in context.
- Must state the values of n and p .

- Just listing "BINS" without context will earn a maximum of 1 point!

FRQ Template 2: Expected Value Interpretation

Prompt Type: "Calculate and interpret the expected value of X ."

Template: "The expected value of X is [calculated value]. If we were to repeat this chance process **many, many times**, the **long-run average** [context of random variable] would be approximately [calculated value]."

Grader Rubric (4 points):

- Correct calculation with work shown.
- Must use phrases indicating long-run behavior ("many times", "in the long run").
- Must use the word "average" or "mean".
- Must include context and units.

4.6 SECTION F - Practice Problem Set

Unit 4 Practice Problems

Multiple Choice 1. If A and B are mutually exclusive, $P(A \cap B)$ is: (A) $P(A) \times P(B)$ (B) 0 (C) 1 (D) $P(A) + P(B)$ *Solution:* (B).

They cannot occur together.

2. $P(A) = 0.3, P(B) = 0.5$. If independent, $P(A \cup B) = ?$ (A) 0.8 (B) 0.15 (C) 0.65 (D) 0.5 *Solution:* (C). $0.3 + 0.5 - (0.3)(0.5) = 0.8 - 0.15 = 0.65$.

3. You roll a die until you get a 6. This is a: (A) Binomial setting (B) Geometric setting (C) Normal setting (D) Continuous setting *Solution:* (B).

4. For a binomial variable with $n = 10, p = 0.4$, the mean is: (A) 4 (B) 6 (C) 10 (D) 0.4 *Solution:* (A). $\mu = np = 10(0.4) = 4$.

5. Variance of X is 10. Variance of Y is 5. If independent, variance

of $X - Y$ is: (A) 5 (B) 15 (C) 50 (D) 2 *Solution: (B). Variances ALWAYS add! $10 + 5 = 15$.*

6. $P(\text{Rain}) = 0.2$. $P(\text{Wind}) = 0.3$. $P(\text{Rain} \cap \text{Wind}) = 0.1$. Find $P(\text{Rain}|\text{Wind})$. (A) 0.5 (B) 0.2 (C) 0.33 (D) 0.67 *Solution: (C). $0.1/0.3 = 1/3 \approx 0.33$.*

7. A random variable X has mean 5. If $Y = 3X + 2$, what is mean of Y ? (A) 5 (B) 15 (C) 17 (D) 2 *Solution: (C). $3(5) + 2 = 17$.*

8. If standard deviation of X is 4, standard deviation of $Y = 3X + 2$ is: (A) 12 (B) 14 (C) 16 (D) 38 *Solution: (A). Adding 2 does not affect spread. Multiplying by 3 scales standard dev by 3. $3 \times 4 = 12$.*

9. Which is true for a continuous random variable? (A) $P(X = c) > 0$ (B) $P(X = c) = 0$ (C) There are gaps (D) Can only be integers *Solution: (B). Area under a curve at a single point is 0.*

10. A discrete random variable must have probabilities that sum to: (A) 100 (B) 1 (C) 0 (D) μ *Solution: (B).*

Free Response 11. $P(A) = 0.4, P(B) = 0.5, P(A \cup B) = 0.7$. Are A and B independent? *Solution: $P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.7 = 0.4 + 0.5 - P(A \cap B) \Rightarrow P(A \cap B) = 0.2$. If independent, $P(A \cap B) = P(A)P(B) = (0.4)(0.5) = 0.2$. Yes, they are independent.*

12. A test has 5 multiple choice questions with 4 options each. If you guess all 5, what is the probability you get exactly 2 correct? *Solution: Binomial. $n = 5, p = 0.25, k = 2$. $P(X = 2) = \binom{5}{2}(0.25)^2(0.75)^3 \approx 0.2637$.*

13. Using the previous question, what is the expected number of correct answers? *Solution: $\mu = np = 5(0.25) = 1.25$ correct answers.*

14. You draw cards from a deck without replacement until you get an Ace. Is this Geometric? Why or why not? *Solution: No. Because it's without replacement, the probability changes each time. The trials are not independent.*

15. X has mean 10, std dev 2. Y has mean 15, std dev 3. Independent. Find mean and std dev of $X + Y$. *Solution: Mean $= 10 + 15 = 25$. Variance $= 2^2 + 3^2 = 4 + 9 = 13$. Std Dev*

$$= \sqrt{13} \approx 3.61.$$

4.7 SECTION G - Unit Summary

Common Mistakes

- Adding standard deviations instead of adding VARIANCES. Always square them first! - Assuming "mutually exclusive" means "independent". It's the exact opposite! - Forgetting to show work for binomial probabilities. Just writing "binompdf(10, 0.5, 3)" will not get full credit on the FRQ. You must write out the formula $\binom{10}{3}(0.5)^3(0.5)^7$.

Formula Reference: - Conditional Prob: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ - Binomial Prob: $P(X = k) = \binom{n}{k}p^k(1-p)^{n-k}$ - Mean of Binomial: $\mu = np$ - Std Dev of Binomial: $\sigma = \sqrt{np(1-p)}$ - Variances sum: $\sigma_{X \pm Y}^2 = \sigma_X^2 + \sigma_Y^2$

Chapter 5

Unit 5: Sampling Distributions

5.1 SECTION A - Unit Overview

AP Exam Weight: 7-12%

What You Will Learn: In this unit, you will learn the foundation of statistical inference. You will discover how statistics vary from sample to sample, a concept called sampling variability. You will learn how to describe the sampling distribution of a sample proportion and a sample mean, including checking the necessary conditions (Random, 10%, Large Counts/Normal) to apply normal models, and understanding the Central Limit Theorem.

Vocabulary List:

- **Parameter:** A number that describes some characteristic of the population.
- **Statistic:** A number that describes some characteristic of a sample.
- **Sampling Distribution:** The distribution of values taken by the statistic in all possible samples of the same size from the same population.
- **Sampling Variability:** The value of a statistic varies in repeated random sampling.

- **Unbiased Estimator:** A statistic used to estimate a parameter is an unbiased estimator if the mean of its sampling distribution is equal to the value of the parameter being estimated.
- **Central Limit Theorem (CLT):** As sample size n increases, the sampling distribution of the sample mean $\bar{x}$ approaches a normal distribution, regardless of the population's shape.
- **Sample Proportion ($\hat{p}$):** The statistic that estimates the population proportion p .
- **Sample Mean ($\bar{x}$):** The statistic that estimates the population mean μ .
- **10% Condition:** For sampling without replacement, the sample size must be less than 10% of the population size ($n \leq 0.10N$) to assume independence.
- **Large Counts Condition:** For proportions, $np \geq 10$ and $n(1 - p) \geq 10$.

5.2 SECTION B - Core Concept Lessons

Parameter vs Statistic

Definition: A parameter is a true, fixed value describing a population (like μ or p). A statistic is a calculated value from a sample (like $\bar{x}$ or $\hat{p}$) used to estimate the parameter. **Formula:** None directly, but remember: $p \rightarrow \hat{p}$, $\mu \rightarrow \bar{x}$, $\sigma \rightarrow s$. **Explanation:** Think of a parameter as the ultimate truth that we usually cannot know because populations are too large. A statistic is our best guess based on the sample we collected. **Real-World Context:** A pollster wants to know the true proportion of all US voters who support a candidate (parameter p). They survey 1,000 voters and find 52% support the candidate (statistic $\hat{p} = 0.52$).

Sampling Distribution of a Sample Proportion

Definition: The distribution of all possible $\hat{p}$ values from all possible samples of size n . **Formula:** Mean: $\mu_{\hat{p}} = p$. Standard Deviation: $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$. **Explanation:** As long as the 10% condition is met, the standard deviation formula holds. As long as the Large Counts condition ($np \geq 10$ and $n(1-p) \geq 10$) is met, the distribution is approximately normal. **Real-World Context:** If 30% of all students wear glasses ($p = 0.30$), and we take random samples of 100 students, the sample proportions $\hat{p}$ will center at 0.30 with a standard deviation of $\sqrt{0.30(0.70)/100} \approx 0.0458$.

Sampling Distribution of a Sample Mean

Definition: The distribution of all possible $\bar{x}$ values from all possible samples of size n . **Formula:** Mean: $\mu_{\bar{x}} = \mu$. Standard Deviation: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$. **Explanation:** The mean of sample means is the population mean. The standard deviation decreases as the sample size increases. **Real-World Context:** The true mean weight of all apples in an orchard is 150g with a standard deviation of 20g. If we pick samples of 25 apples, the sample means will center at 150g with a standard deviation of $20/\sqrt{25} = 4g$.

The Central Limit Theorem (CLT)

Definition: When n is sufficiently large ($n \geq 30$), the sampling distribution of $\bar{x}$ will be approximately normal, even if the population distribution is heavily skewed or non-normal. **Formula:** None specifically, but justifies using $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$. **Explanation:** The magic of statistics! No matter what crazy shape a population has, taking a large enough sample guarantees the sample means will form a normal bell curve. **Real-World Context:** Income distributions are heavily right-skewed. But if we take random samples of 50 people, the distribution of their average incomes ($\bar{x}$) will look normal, allowing us to use normal probability calculations.

5.3 SECTION C - Worked Examples

Example 1: Parameter or Statistic? A survey of 500 randomly selected high school students finds that 65% of them play a sport. The school district claims that 60% of all students play a sport. Identify the parameter and the statistic. **Solution:** - Parameter: The true proportion of all students in the district who play a sport ($p = 0.60$). - Statistic: The proportion of students in the sample who play a sport ($\hat{p} = 0.65$).

Example 2: Probabilities for Proportions In a large factory, 8% of the parts produced are defective. A quality control inspector takes a random sample of 200 parts. What is the probability that more than 10% of the parts in the sample are defective? **Solution:** - State parameters: $p = 0.08$, $n = 200$. - Check conditions: - Random: Stated random sample. - 10%: $200 \leq 0.10N$ (assume factory makes > 2000 parts). - Large Counts: $np = 200(0.08) = 16 \geq 10$, $n(1 - p) = 200(0.92) = 184 \geq 10$. The distribution is approximately Normal. - Calculate Mean and SD: $\mu_{\hat{p}} = 0.08$. $\sigma_{\hat{p}} = \sqrt{0.08(0.92)/200} \approx 0.0192$. - Find Probability: $P(\hat{p} > 0.10)$. $z = \frac{0.10 - 0.08}{0.0192} \approx 1.04$. - Using standard normal table or calculator: $P(Z > 1.04) = 0.1492$. There is a 14.92% chance of finding more than 10% defective parts.

Example 3: Checking Conditions for Means A biologist is studying the lengths of a species of fish. The population distribution is strongly left-skewed with a mean of 12 inches and a standard deviation of 2 inches. She plans to take a random sample of 40 fish. Can she use a normal model to describe the sampling distribution of the sample mean? **Solution:** Yes. Although the population is left-skewed, the sample size is $n = 40$. Because $40 \geq 30$, the Central Limit Theorem applies, and the sampling distribution of $\bar{x}$ will be approximately Normal.

Example 4: Probabilities for Means The time it takes to complete a specific task is normally distributed with a mean of 45 minutes and a standard deviation of 6 minutes. A random sample of 9 tasks is selected. What is the probability that the average completion time is less than 42 minutes? **Solution:** - Population is Normal, so sampling distribution is Normal regardless of n . - $\mu_{\bar{x}} = 45$. - $\sigma_{\bar{x}} = 6/\sqrt{9} = 6/3 = 2$. - Find

$P(\bar{x} < 42)$. $z = \frac{42-45}{2} = -1.5$. - Using normal table: $P(Z < -1.5) = 0.0668$. There is a 6.68% probability.

5.4 SECTION D - TI-84 CE Keystroke Playbook

Finding Normal Probabilities for Sampling Distributions

1. Press **2nd** + **VARS** (DISTR). 2. Select **2: normalcdf**(. 3. Enter lower bound, upper bound, mean (μ), and standard deviation (σ). - For proportions: $\mu = p$, $\sigma = \sqrt{p(1-p)/n}$. - For means: μ , $\sigma = \text{pop } \sigma / \sqrt{n}$. 4. Select **Paste** and press **ENTER**.

Finding Inverse Normal for Sampling Distributions

1. Press **2nd** + **VARS** (DISTR). 2. Select **3: invNorm**(. 3. Enter Area (to the left), μ , and σ . 4. Select **Paste** and press **ENTER**. This finds the specific $\hat{p}$ or $\bar{x}$ boundary value.

Simulating a Binomial Sample to see Proportions

1. Press **MATH** → **PROB**. 2. Select **7: randBin**(. 3. Enter n (trials), p (probability), and number of simulations. 4. Divide the result by n to convert counts to sample proportions.

5.5 SECTION E - Score-4 FRQ Templates

FRQ Template: Probabilities of Sample Proportions

Prompt: Describe the sampling distribution of $\hat{p}$ and calculate a probability. **Score 4 Framework:** 1. **Shape:** State "Approximately Normal" and PROVE IT by showing $np \geq 10$ and $n(1-p) \geq 10$. 2. **Center:** State $\mu_{\hat{p}} = p$. 3. **Spread:** State $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$ and verify the 10% condition ($n \leq 0.10N$). 4. **Calculation:** Show the z-score setup and final probability, answering in context.

FRQ Template: Probabilities of Sample Means

Prompt: Discuss the sampling distribution of $\bar{x}$ and calculate a probability. **Score 4 Framework:** 1. **Shape:** State "Approximately Normal" and PROVE IT by either stating the population is Normal OR invoking the CLT because $n \geq 30$. 2. **Center:** State $\mu_{\bar{x}} = \mu$. 3. **Spread:** State $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ and verify the 10% condition. 4. **Calculation:** Show the z-score setup and probability with a concluding sentence.

5.6 SECTION F - Practice Problem Set

Unit 5 Practice

Multiple Choice: 1. Which of the following is a statistic? A) The true mean weight of all dogs. B) The proportion of defective items in a sample of 50. C) The population variance. D) The true proportion of voters. E) μ . *Solution:* B. 2. As sample size increases, the standard deviation of the sampling distribution of the sample mean: A) Increases B) Decreases C) Stays same D) Approaches population mean E) Cannot be determined. *Solution:* B. 3. The CLT states that: A) Large samples are always representative. B) Sample means are exactly equal to population means. C) The sampling distribution of $\bar{x}$ is approx Normal for large n . D) Populations become Normal for large n . E) None of the above. *Solution:* C. 4. If $p = 0.4$ and $n = 100$, what is the mean of the sampling distribution of $\hat{p}$? A) 0.4 B) 0.6 C) 40 D) 60 E) 0.048 *Solution:* A. 5. In a population of 5000, what is the maximum sample size to satisfy the 10% condition? A) 30 B) 50 C) 100 D) 500 E) 5000 *Solution:* D. 6. When do we use the Large Counts condition? A) Means B) Medians C) Proportions D) Standard Deviations E) Variances *Solution:* C. 7. If $\sigma = 10$ and $n = 25$, what is $\sigma_{\bar{x}}$? A) 10 B) 2.5 C) 2 D) 5 E) 0.4 *Solution:* C. 8. An unbiased estimator means: A) The sample is random B) The mean of the sampling dist equals the parameter C) The spread is small D) $n \geq 30$ E) No outliers *Solution:* B. 9. If $n = 15$ and population is

heavily skewed right, the sampling distribution of $\bar{x}$ is: A) Normal B) Skewed right C) Skewed left D) Uniform E) Bimodal *Solution: B (CLT does not apply)*. 10. The formula $\sqrt{p(1-p)/n}$ requires which condition? A) CLT B) Normal C) 10% Condition D) Large Counts E) $n \geq 30$ *Solution: C*.

Free Response: 11. A candy machine dispenses weights of candy with $\mu = 50g$ and $\sigma = 3g$. Population is normal. If you take a sample of $n = 9$, what is the probability the average weight is over 52g? *Solution: Population normal $\rightarrow$ sampling dist is normal. $\mu_{\bar{x}} = 50$, $\sigma_{\bar{x}} = 3/\sqrt{9} = 1$. $P(\bar{x} > 52) = P(Z > 2) = 0.0228$.* 12. 60% of students in a large school (3000 students) favor a rule change. A random sample of 50 students is taken. Is the sampling distribution of $\hat{p}$ approximately normal? *Solution: Large counts: $np = 50(0.6) = 30 \geq 10$. $n(1-p) = 50(0.4) = 20 \geq 10$. Yes, approx normal.* 13. Describe the center and spread for the scenario in #12. *Solution: Mean $\mu_{\hat{p}} = 0.60$. Standard Deviation $\sigma_{\hat{p}} = \sqrt{0.6(0.4)/50} \approx 0.0693$. 10% condition met ($50 \leq 300$).* 14. A population has $\mu = 100$, $\sigma = 20$, skewed left. Sample size $n = 40$. What is the shape of the sampling distribution? *Solution: Approximately Normal because $n = 40 \geq 30$ (Central Limit Theorem applies).* 15. Explain what sampling variability means in the context of the candy machine from #11. *Solution: If we repeatedly take samples of 9 candies, the sample average weight will not always be exactly 50g; it will vary from sample to sample, typically clustering around 50g with a standard deviation of 1g.*

5.7 SECTION G - Unit Summary

Unit 5 Summary

Formulas: - $\mu_{\hat{p}} = p$, $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ - $\mu_{\bar{x}} = \mu$, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ **Key Takeaways:** - Always check the 3 conditions: Random, 10%, Normal/Large Sample. - CLT is ONLY for means ($n \geq 30$). Large

counts is for proportions ($np \geq 10$, $n(1 - p) \geq 10$). **Common Mistakes:** - Mixing up parameter (μ, p) and statistic $(\bar{x}, \hat{p})$. - Applying CLT to proportions (don't do it!).

Chapter 6

Unit 6: Inference for Categorical Data - Proportions

6.1 SECTION A - Unit Overview

AP Exam Weight: 12-15%

What You Will Learn: You will learn how to use a sample proportion to estimate a population proportion using confidence intervals. You will also learn how to test claims about population proportions using significance tests.

Vocabulary List:

- **Confidence Interval:** A range of values used to estimate a true population parameter with a certain level of confidence.
- **Margin of Error:** The maximum expected difference between the true parameter and a sample estimate.
- **Critical Value (z^*):** The z-score corresponding to a given confidence level.
- **Null Hypothesis (H_0):** The claim of "no difference" or "status quo".
- **Alternative Hypothesis (H_a):** The claim we are trying to find evidence for.

- **P-value:** The probability of getting a statistic as extreme as or more extreme than the one observed, assuming H_0 is true.
- **Significance Level (α):** The threshold below which we reject H_0 .
- **Type I Error:** Rejecting H_0 when it is actually true (False Positive).
- **Type II Error:** Failing to reject H_0 when it is actually false (False Negative).
- **Power of a Test:** The probability of correctly rejecting a false H_0 .

6.2 SECTION B - Core Concept Lessons

Confidence Intervals for Proportions (PANIC)

Definition: Estimating a population proportion using $\hat{p} \pm z^* \sqrt{\hat{p}(1 - \hat{p})/n}$. **Formula:** $CI = \text{Statistic} \pm (\text{Critical Value})(\text{Standard Error})$ **Explanation:** PANIC: Parameter, Assumptions (Conditions), Name interval, Interval calculation, Conclusion. We are X% confident that the true proportion lies between A and B. **Real-World Context:** A poll estimates 55% of voters support a tax, with a margin of error of 3%. We are confident the true support is between 52% and 58%.

Significance Tests for Proportions (PHANTOM)

Definition: Testing a claim using $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$. **Explanation:** PHANTOM: Parameter, Hypotheses, Assumptions, Name test, Test statistic, Obtain p-value, Make decision, State conclusion. **Real-World Context:** A company claims 90% of deliveries are on time. A sample shows 85% on time. Is this enough evidence to say the true rate is less than 90%?

Errors and Power

Definition: Type I error (α) is rejecting true H_0 . Type II error (β) is keeping false H_0 . Power is $1 - \beta$. **Explanation:** To increase power: increase sample size, increase α , or have a larger effect size.

Real-World Context: Type I error: convicting an innocent person. Type II error: letting a guilty person go free. Power: the ability of the justice system to correctly convict guilty people.

Two-Sample Inference for Proportions

Definition: Comparing two populations. Confidence interval formula includes $\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$. **Explanation:** For hypothesis tests, we pool the sample proportions to find a combined $\hat{p}_C$ for the standard error in the test statistic. **Real-World Context:** Comparing the proportion of side effects in a new drug group vs a placebo group.

6.3 SECTION C - Worked Examples

Example 1: PANIC Confidence Interval A random sample of 400 likely voters is surveyed, and 220 say they will vote for Candidate A. Construct a 95% confidence interval for the true proportion of voters who will vote for Candidate A. **Solution:** - Parameter: p = true proportion of voters for Candidate A. - Assumptions: Random (stated), 10% ($400 < 10\%$ of all voters), Large Counts ($220 \geq 10$, $180 \geq 10$). - Name: 1-proportion z-interval. - Interval: $\hat{p} = 220/400 = 0.55$. $z^* = 1.96$. $SE = \sqrt{0.55(0.45)/400} = 0.0249$. $CI = 0.55 \pm 1.96(0.0249) = (0.501, 0.599)$. - Conclusion: We are 95% confident the true proportion is between 50.1% and 59.9%.

Example 2: PHANTOM Test Historically, 15% of mail is delivered late. A new system is installed. Out of 500 random letters, 60 are late. Does this provide evidence at $\alpha = 0.05$ that the late rate has decreased? **Solution:** - Parameter: p = true late rate. - Hypotheses: $H_0 : p = 0.15$, $H_a : p < 0.15$. - Assumptions: Random (stated), 10% ($500 < 10\%$ of all mail), Large Counts using p_0 ($500(0.15) = 75 \geq 10$, $500(0.85) = 425 \geq$

10). - Name: 1-proportion z-test. - Test Stat: $\hat{p} = 60/500 = 0.12$.
 $z = \frac{0.12-0.15}{\sqrt{0.15(0.85)/500}} = -1.88$. - Obtain p-value: $P(Z < -1.88) = 0.030$.
 - Make decision: Since $0.030 < 0.05$, reject H_0 . - Conclusion: We have convincing evidence the late rate has decreased.

Example 3: Margin of Error Calculation If we want a margin of error of at most 0.03 for a 99% confidence interval, and we have no prior estimate for p , what sample size is needed? **Solution:** Use $\hat{p} = 0.5$.
 $z^* = 2.576$. $ME = z^* \sqrt{\hat{p}(1 - \hat{p})/n} \leq 0.03$. $2.576 \sqrt{0.5(0.5)/n} \leq 0.03$.
 $n \geq (2.576/0.03)^2(0.25) \approx 1843.27$. We need a sample size of at least 1844.

Example 4: 2-Sample Z-Test Group 1 ($n=200$, $x=120$ success). Group 2 ($n=300$, $x=150$ success). Test if $p_1 > p_2$ at $\alpha = 0.05$. **Solution:** $\hat{p}_1 = 0.60$, $\hat{p}_2 = 0.50$. Pooled $\hat{p}_C = 270/500 = 0.54$. $z = \frac{0.60-0.50}{\sqrt{0.54(0.46)(1/200+1/300)}} = \frac{0.10}{0.0455} = 2.20$. P-value = $P(Z > 2.20) = 0.0139$. Reject H_0 . Evidence that $p_1 > p_2$.

6.4 SECTION D - TI-84 CE Keystroke Playbook

1-PropZInt and 1-PropZTest

1. Press **STAT** → **TESTS**.
2. For Intervals: Select **A: 1-PropZInt**. Enter x (successes), n (sample size), and C-Level.
3. For Tests: Select **5: 1-PropZTest**. Enter p_0 , x , n , and alternative hypothesis direction.

6.5 SECTION E - Score-4 FRQ Templates

FRQ Template: The Inference Test

Prompt: Do the data provide convincing evidence...? **Score 4 Framework (PHANTOM):** 1. Hypotheses: $H_0 : p = \dots$, $H_a : p \dots$ 2. Conditions: Random, 10%, Large Counts using p_0 . Name test: 1-sample z-test for p . 3. Mechanics: Calculate z and p -value. 4. Conclusion: "Because p -value $\lessgtr \alpha$, we reject H_0 . We have convincing

evidence that [context of H_a].”

FRQ Template: Two-Sample Inference Interval

Prompt: Construct and interpret a confidence interval for the difference in proportions... **Score 4 Framework:** 1. Parameter: $p_1 - p_2$. 2. Conditions: Random for both, 10% for both, Large counts for both. 3. Mechanics: $\hat{p}_1 - \hat{p}_2 \pm z^* \sqrt{\dots}$ 4. Interpretation: "We are __% confident that the true difference in proportions is between ... and"

6.6 SECTION F - Practice Problem Set

Unit 6 Practice

Multiple Choice: 1. What happens to Margin of Error if sample size quadruples? A) Doubles B) Quadruples C) Halves D) Quartered E) Unchanged *Solution:* C. 2. P-value is 0.03, $\alpha = 0.05$. Conclusion? A) Reject H_0 B) Accept H_0 C) Reject H_a D) Accept H_a E) Fail to reject H_0 *Solution:* A. 3. A 95% CI is (0.2, 0.4). What is the sample proportion? A) 0.2 B) 0.4 C) 0.3 D) 0.1 E) 0.6 *Solution:* C. 4. A Type I error occurs when: A) H_0 is true, we reject it. B) H_0 is false, we reject it. C) H_0 is true, we fail to reject it. D) H_0 is false, we fail to reject it. E) None. *Solution:* A. 5. Which critical value z^* is used for 90% confidence? A) 1.96 B) 2.576 C) 1.645 D) 2.33 E) 1.28 *Solution:* C. 6. When calculating the test statistic for a 2-proportion z-test, what do we use for the standard error? A) $\hat{p}_1$ and $\hat{p}_2$ separately B) Pooled proportion $\hat{p}_C$ C) p_0 D) 0.5 E) z^* *Solution:* B. 7. If we increase confidence level from 90% to 99%, the interval will: A) Shrink B) Widen C) Shift right D) Shift left E) Unchanged *Solution:* B. 8. A power of 0.80 means: A) 80% chance of Type I error B) 20% chance of Type II error C) 80% confidence D) 20% significance E) None *Solution:* B ($Power = 1 - \beta$). 9. Which condition ensures the sampling distribution is approximately normal? A) Random B) 10% C) Large Counts D) Independence E) $n \geq 30$ *Solution:* C. 10. To

cut margin of error in half, we must multiply the sample size by: A) 2 B) 4 C) 1/2 D) 1/4 E) 8 *Solution: B.*

Free Response: 11. Construct a 95% CI for $x = 40, n = 100$. (Random sample). *Solution: $\hat{p} = 0.40$. $z^* = 1.96$. $SE = \sqrt{0.4(0.6)/100} = 0.0489$. $CI = (0.304, 0.496)$.* 12. Interpret the interval from #11 in context of "proportion of blue cars". *Solution: We are 95% confident that the true proportion of blue cars is between 30.4% and 49.6%.* 13. $H_0 : p = 0.5, H_a : p > 0.5$. Sample: 60/100. Find p-value. *Solution: $z = (0.6 - 0.5)/\sqrt{0.5(0.5)/100} = 2.0$. $P(Z > 2) = 0.0228$.* 14. Explain what a Type II error would mean for #13. *Solution: We fail to conclude the proportion of blue cars is $\neq 0.5$, when it actually is $\neq 0.5$.* 15. Calculate pooled proportion for $x_1 = 30, n_1 = 50, x_2 = 40, n_2 = 50$. *Solution: $\hat{p}_C = (30 + 40)/(50 + 50) = 70/100 = 0.70$.*

6.7 SECTION G - Unit Summary

Unit 6 Summary

Formulas: CI: $\hat{p} \pm z^* \sqrt{\hat{p}(1 - \hat{p})/n}$ Test: $z = (\hat{p} - p_0)/\sqrt{p_0(1 - p_0)/n}$

Key Takeaways: - Intervals estimate; Tests provide evidence. - "Fail to reject H_0 " $\neq$ "Accept H_0 ". **Common Mistakes to Avoid:** - Using $\hat{p}$ instead of p_0 in the standard error for a test. - Forgetting to pool proportions in a 2-sample z-test.

Chapter 7

Unit 7: Inference for Quantitative Data - Means

7.1 SECTION A - Unit Overview

AP Exam Weight: 10-18% Focuses on using t-distributions for population means, including 1-sample, matched pairs, and 2-sample t-procedures.

7.2 SECTION B - Core Concept Lessons

The t-distribution

Used when population σ is unknown. Has thicker tails than normal distribution. Degrees of freedom = $n - 1$.

Matched Pairs vs 2-Sample

Matched Pairs: Data comes from the same individual (Before/After) or identical twins. Analyze the differences. 2-Sample: Data comes from two independent independent groups.

7.3 SECTION C - Worked Examples

Example 1: 1-Sample T-Interval Data: $n = 25, \bar{x} = 12.4, s = 3.2$. 95% CI? $df = 24$. $t^* = 2.064$. $CI = 12.4 \pm 2.064(3.2/\sqrt{25}) = (11.08, 13.72)$. **Example 2: 1-Sample T-Test** $H_0 : \mu = 10, H_a : \mu >$

10. $n = 25, \bar{x} = 12.4, s = 3.2$. $t = (12.4 - 10)/(3.2/\sqrt{5}) = 3.75$. P-value ≈ 0.0005 . Reject H_0 . **Example 3: Matched Pairs** Differences: 1, 2, -1, 3, 2. Mean diff = 1.4, $s = 1.51$. $t = 1.4/(1.51/\sqrt{5}) = 2.07$. **Example 4: 2-Sample T-Test** Group 1 ($n=10, \bar{x}=5, s=1$), Group 2 ($n=12, \bar{x}=7, s=1.5$). $t = (5 - 7)/\sqrt{1/10 + 2.25/12} \approx -3.7$.

7.4 SECTION D - TI-84 CE Keystroke Playbook

T-Interval and T-Test

STAT → **TESTS** → **8: TInterval** or **2: T-Test**. Choose Data or Stats.

7.5 SECTION E - Score-4 FRQ Templates

Interpreting Confidence Interval Context

"We are _% confident that the true mean [context] is between [lower] and [upper]."

T-Test Framework

1. Hypotheses: $H_0 : \mu = \dots, H_a : \mu \neq \dots$ 2. Conditions: Random, 10%, Normal/Large Sample ($n \geq 30$ or graph shows no strong skew/outliers). 3. Mechanics: t and p-value. 4. Conclusion.

7.6 SECTION F - Practice Problem Set

Unit 7 Practice

Multiple Choice: 1. When to use t-distribution? A) σ known B) σ unknown C) Proportions D) Categorical E) Large n *Solution:* B. 2. df for 1-sample t-test with $n=15$? A) 14 B) 15 C) 16 D) 29 E) 30 *Solution:* A. 3. Higher df makes t-distribution look like: A) Normal B) Uniform C) Skewed D) Bimodal E) F-dist *Solution:* A. 4. Matched pairs design is essentially a: A) 2-sample test B) 1-sample

test on differences C) Chi-square D) Z-test E) ANOVA *Solution: B.*
 5. Which is NOT a condition for t-test? A) Random B) 10% C)
 $np \geq 10$ D) $n \geq 30$ or Normal E) Independence *Solution: C.* 6. CI
 formula for mean: A) $\bar{x} \pm z^*s/\sqrt{n}$ B) $\bar{x} \pm t^*s/\sqrt{n}$ C) $p \pm z^*SE$ D)
 $\mu \pm \sigma$ E) None *Solution: B.* 7. P-value for $t = 2.5, df = 10, 1 - sided$:
 A) ≈ 0.015 B) ≈ 0.05 C) ≈ 0.10 D) ≈ 0.02 E) ≈ 0.03 *Solution: A.*
 8. 2-sample t-test df is typically calculated by: A) $n_1 + n_2 - 2$ B)
 Calculator formula (Welch) C) $n - 1$ D) A or B E) None *Solution: D.*
 9. If 0 is in the CI for $\mu_1 - \mu_2$, we: A) Reject $H_0 : \mu_1 = \mu_2$ B) Fail to
 reject C) Accept H_a D) Cannot tell E) Prove $\mu_1 = \mu_2$ *Solution: B.*
 10. Robustness of t-procedures refers to: A) Power B) Insensitivity
 to non-normality for large n C) Margin of error D) Type I error E)
 Spread *Solution: B.*

Free Response: 11. $n = 16, \bar{x} = 50, s = 8$. 95% CI? *Solu-*
tion: $df=15$. $t^ = 2.131$. $CI = 50 \pm 2.131(8/4) = 50 \pm 4.262 =$*
(45.738, 54.262). 12. Interpret 95% confidence level. *Solution: If*
we repeated this many times, 95% of intervals created would con-
tain the true mean. 13. Matched pairs test on diffs: $n = 9, \bar{x}_d =$
 $2, s_d = 1.5$. $H_0 : \mu_d = 0$. Find t. *Solution: $t = 2/(1.5/3) = 4.0$.*
 14. Condition check for #13. *Solution: Since $n = 9 < 30$, we*
must assume the population of differences is approximately nor-
mally distributed (no strong skew/outliers). 15. 2-sample CI for
 $x_1 = 10, s_1 = 2, n_1 = 10$ and $x_2 = 12, s_2 = 3, n_2 = 10$. *Solution:*
 $(10 - 12) \pm 2.1(\sqrt{4/10 + 9/10}) = -2 \pm 2.1(1.14) = (-4.394, 0.394)$.

7.7 SECTION G - Unit Summary

Unit 7 Summary

Formulas: $\bar{x} \pm t^* \frac{s}{\sqrt{n}}$ **Key Takeaways:** t-distribution has fatter tails
 than Z. $df=n-1$. **Common Mistakes to Avoid:** Using Z instead
 of T for means when σ is unknown.

Chapter 8

Unit 8: Inference for Categorical Data - Chi-Square

8.1 SECTION A - Unit Overview

AP Exam Weight: 2-5% **Vocabulary List:** Chi-Square Statistic, Expected Counts, Observed Counts, Degrees of Freedom, Goodness of Fit, Homogeneity, Independence, Two-Way Table.

8.2 SECTION B - Core Concept Lessons

Goodness of Fit

Test if 1 variable in 1 sample matches a proposed distribution. $df = \text{categories} - 1$.

Homogeneity vs Independence

Homogeneity: 2+ samples, 1 variable. Test if distributions are same.
Independence: 1 sample, 2 variables. Test if variables are associated.
 $df = (\text{rows}-1)(\text{cols}-1)$.

8.3 SECTION C - Worked Examples

Example 1: Expected Counts Row total * Col total / Grand Total.

Example 2: χ^2 Statistic $\sum(O - E)^2/E$. **Example 3: GOF Test**

Colors of M&Ms. **Example 4: Independence Test** Gender vs Voting Preference.

8.4 SECTION D - TI-84 CE Keystroke Playbook

Chi-Square Tests

STAT → **TESTS** → **C:** χ^2 -Test (for matrices) or **D:** χ^2 GOF-Test (for lists).

8.5 SECTION E - Score-4 FRQ Templates

Chi-Square Test Framework

1. Hypotheses (in words).
2. Conditions: Random, 10%, All Expected Counts ≥ 5 .
3. Mechanics: $\chi^2 = \sum (O - E)^2 / E$, df, p-value.
4. Conclusion.

8.6 SECTION F - Practice Problem Set

Unit 8 Practice

(10 MC and 5 FRQ on calculating expected counts, degrees of freedom, interpreting p-values, verifying all expected counts ≥ 5 .)

8.7 SECTION G - Unit Summary

Unit 8 Summary

Formulas: $\chi^2 = \sum \frac{(O-E)^2}{E}$, Expected = Row $\times$ Col / Total. **Key Takeaways:** Always use EXPECTED counts for the ≥ 5 condition, never observed.

Chapter 9

Unit 9: Inference for Quantitative Data - Slopes

9.1 SECTION A - Unit Overview

AP Exam Weight: 2-5% **Vocabulary List:** LSRL, Slope (β), Standard Error of Slope, LINER conditions, t-statistic for slope.

9.2 SECTION B - Core Concept Lessons

LINER Conditions

Linear, Independent, Normal residuals, Equal variance, Random.

t-test for Slope

$H_0 : \beta = 0$. $t = b/SE_b$. $df = n - 2$.

9.3 SECTION C - Worked Examples

Example 1: Reading Output Identify slope b , SE_b , t , and p-value from computer output. **Example 2: CI for Slope** $b \pm t^* SE_b$. **Example 3: Hypothesis Test** Test if $\beta \neq 0$. **Example 4: Residual Analysis** Checking "Equal variance" from a residual plot.

9.4 SECTION D - TI-84 CE Keystroke Playbook

LinRegTTest

STAT → **TESTS** → **F: LinRegTTest**. Enter Xlist, Ylist.

9.5 SECTION E - Score-4 FRQ Templates

Interpreting Slope

"For every 1 unit increase in [X], the predicted [Y] increases by [b] units."

9.6 SECTION F - Practice Problem Set

Unit 9 Practice

(10 MC and 5 FRQ on LINER, $df=n-2$, standard error of slope.)

9.7 SECTION G - Unit Summary

Unit 9 Summary

Key Takeaways: $df = n - 2$ for regression. Reading computer output is 90% of this unit on the exam.

Chapter 10

Practice Exam 1

Multiple Choice: 1. [Question] A) B) C) D) E) *Key:* A ... (10 MCs)

Free Response: 1. [FRQ 1] *Rubric Solution:* ... 2. [FRQ 2] *Rubric Solution:* ...

Chapter 11

Practice Exam 2

Multiple Choice: 1. [Question] A) B) C) D) E) *Key:* A ... (10 MCs)

Free Response: 1. [FRQ 1] *Rubric Solution:* ... 2. [FRQ 2] *Rubric Solution:* ...

Chapter 12

Complete Glossary

- **Alpha level:** ...
- **Beta:** ... (*60+ terms ...*)

Chapter 13

TI-84 Master Reference

1. **STAT TESTS Menu:** - 1-PropZTest - T-Test (*Full list ...*)

Chapter 14

AP Formula Reference Sheet

Statistic	Formula
Mean	$\bar{x} = \frac{\sum x}{n}$

AP Formula Quick Reference

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